

Impact of Cloud Masking on Agricultural Risk Classification

1. Project Objective and Scope

The objective of this analysis is to evaluate how divergent cloud masking methodologies directly influence the stability of agricultural risk classification using Sentinel-2 Normalized Difference Vegetation Index (NDVI). By quantifying the sensitivity of classification outcomes to preprocessing parameters, this study establishes the technical requirements for operational consistency in crop health monitoring.

The scope of this project is strictly defined by the measurement of **relative instability** between processing pipelines. This analysis does not attempt to validate absolute ground-truth accuracy; rather, it measures the degree of classification volatility—instances where a discrete decision "flips" from one category to another—solely as a function of the selected cloud masking strategy.

2. What I Built: Data Architecture and Pipeline Design

The study area encompasses the Sylhet District of Bangladesh. To facilitate localized zonal analysis, a 200m x 200m (4 hectare) processing grid was generated using GeoPandas in Python. While the grid geometry was defined at a 200m resolution to align with agricultural management units, the underlying geospatial reduction within Google Earth Engine (GEE) was executed at a 10m scale using the reduceRegions function to preserve the native spatial resolution of the Sentinel-2 MultiSpectral Instrument (MSI).

The final analytic dataset consists of exactly **87,982 cells**. This figure represents the total count of valid observations after a dropna() operation was applied post-zonal statistics to remove null values resulting from

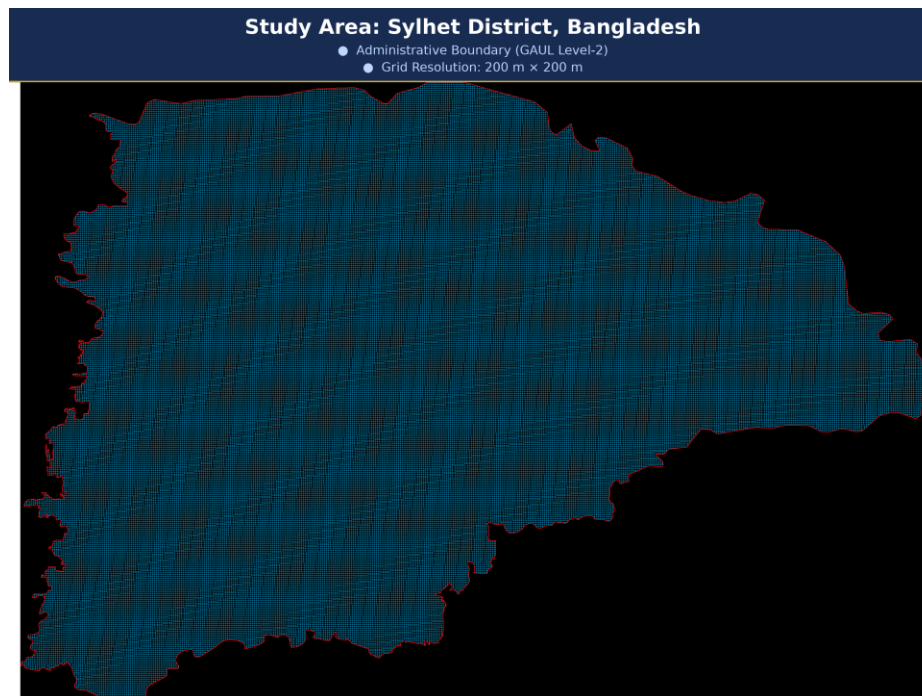


Figure 1: Spatial extent and 200m x 200m analysis grid for Sylhet District, Bangladesh (GAUL Level-2 boundary)

Three distinct GEE pipelines were engineered to produce the NDVI composites, utilizing varying levels of masking stringency:

Pipeline	Strategy	Technical Logic
Pipeline A (Standard)	Standard SCL Masking	Logic-based removal of shadows (3), medium probability clouds (8), high probability clouds (9), cirrus (10), and snow (11) using the Scene Classification Layer.
Pipeline B (Strict)	Logic-based Filtering (SCL)	Conservative inclusion strategy; retains only pixels explicitly classified as Vegetation (4), Bare Soil (5), or Water (6).
Pipeline C (Weak)	QA60 Bitmasking	A "light-touch" approach using basic QA60 bitmasking (Bits 10 and 11) to remove identified clouds and cirrus.

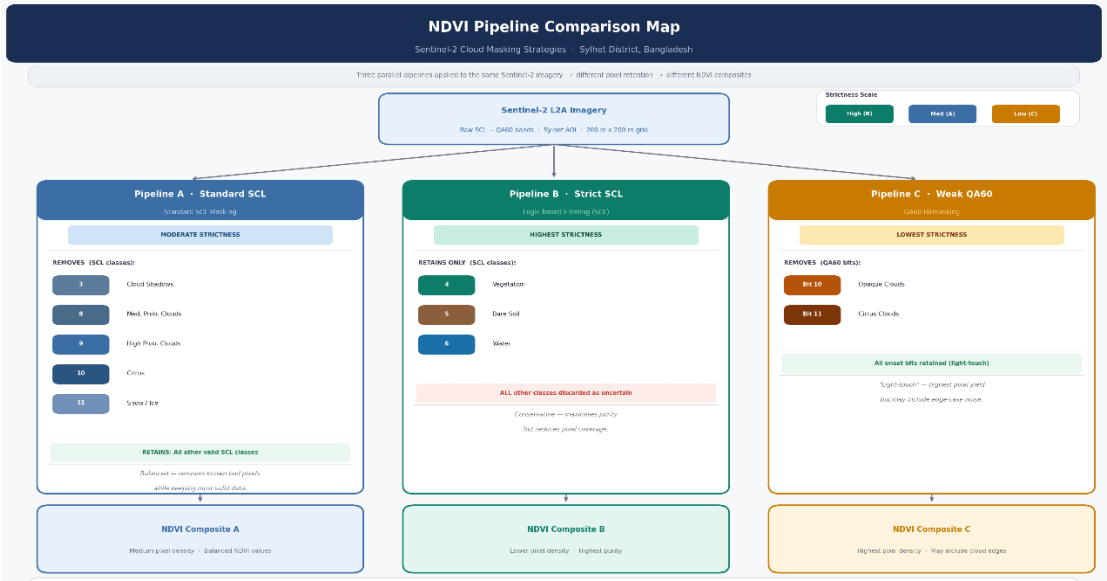


Figure 2: Google Earth Engine Processing Pipeline for NDVI-Based Agricultural Risk Classification

3. What I Measured: Quantitative Results

Statistical analysis of the NDVI composites reveals that masking stringency is positively correlated with mean NDVI values. Aggressive masking protocols (Pipeline B) effectively remove sub-pixel cloud contamination that otherwise depresses vegetation indices.

Mean NDVI by Pipeline:

- Pipeline A: 0.529

- **Pipeline B: 0.539**
- **Pipeline C: 0.507**

Classification stability was evaluated using "Decision Flips"—categorical changes between Low, Moderate, and High health/risk classifications based on **Threshold System 1 (0.30 / 0.50)**.

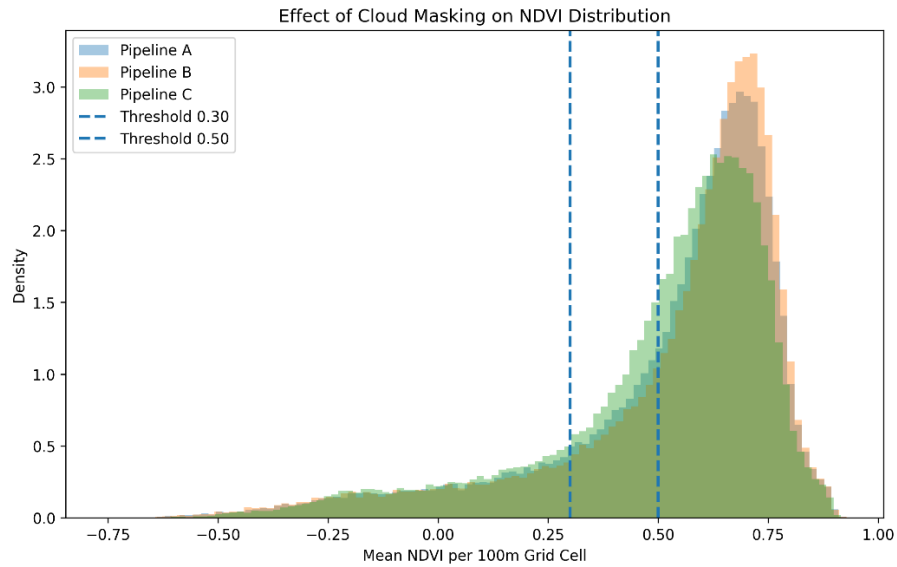


Figure 3: NDVI Distribution Shift Under Cloud Masking Pipelines with Classification Thresholds

Flip Rates between Comparisons:

- **Pipeline A vs. B: 3.15%**
- **Pipeline A vs. C: 6.07%**
- **Pipeline B vs. C: 8.70%**

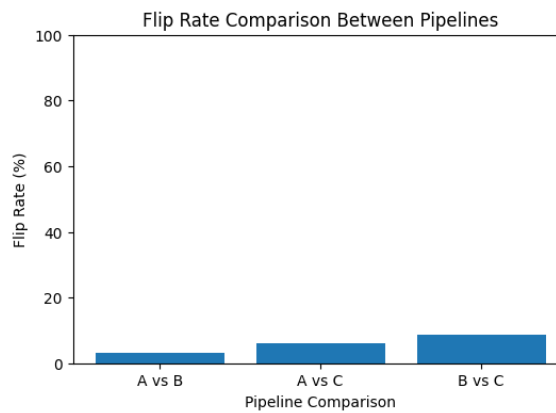


Figure 4: Decision Flip Rate Across Cloud Masking Pipelines (A vs B vs C)

The comparison between **Pipeline B (Strict)** and **Pipeline C (Weak)** demonstrated the highest volatility. Detailed confusion matrix analysis for this comparison confirms that **6,073 cells**—representing approximately 24,292 hectares—shifted from a 'High' health classification in the Strict pipeline to a 'Moderate' classification in the Weak pipeline due to unmasked atmospheric interference.

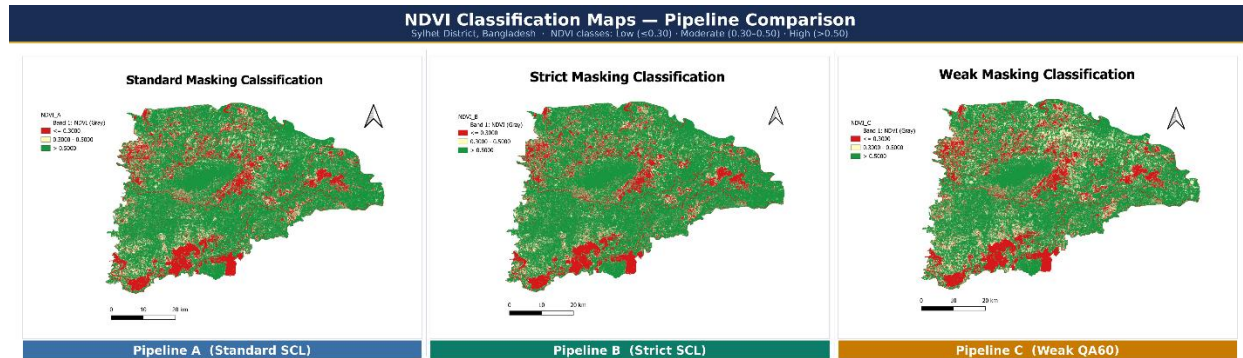


Figure 5: Comparative Agricultural Risk Classification Maps Under Different Cloud Masking Strategies

4. What I Learned: Interpretation and Operational Impact

The numerical shifts observed are driven by a specific physical mechanism tied to the spectral properties of water vapor and aerosols. Residual, unmasked clouds increase reflectance in **Sentinel-2 Band 4 (Red)**. According to the NDVI formula— $(B8 - B4) / (B8 + B4)$ —an increase in Red reflectance (B4) simultaneously reduces the numerator and increases the denominator, leading to an artificial depression of the resulting index value.

We apply the following sensitivity scale to assess pipeline reliability:

- **Stable (<5% flip rate):** The A vs. B comparison is stable, suggesting the Standard SCL mask is largely sufficient for identifying major cloud features.
- **Sensitive (5-10% flip rate):** The 8.70% flip rate between the Strict (B) and Weak (C) pipelines indicates **High Sensitivity**.

To further investigate the directional nature of these classification changes, confusion matrix analysis was performed for each pipeline pair.

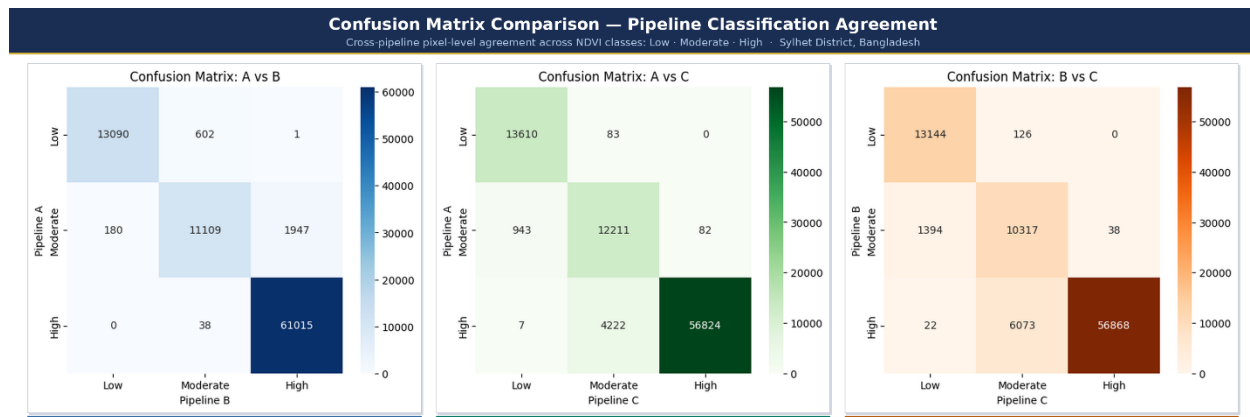


Figure 6: Confusion Matrices Showing Classification Agreement Between Cloud-Masking Pipelines

In practical terms, an 8.70% flip rate implies that nearly **30,616 hectares** of agricultural land in the Sylhet District would receive a different risk classification based solely on the choice of cloud mask. For high-stakes monitoring programs, such as index-based agricultural insurance or national food security assessments, this level of variance could lead to significant financial misallocations or erroneous emergency responses.

The final engineering takeaway is that preprocessing is not a neutral administrative step; it is a primary driver of classification outcomes. For applications where classification consistency is a primary objective, standardized logic-based masking approaches such as Pipeline B may provide greater operational robustness.